Vahid Arastuian

Applied Geospatial & Computer Vision Engineer

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Summary

Applied Geospatial and Computer Vision Engineer with an MSc in Photogrammetry and 7+ years of experience in UAV imagery, GIS, remote sensing, and deep learning-based perception systems. Specialized in developing end-to-end pipelines for photogrammetry, object detection, change detection, and 3D reconstruction using Python, PyTorch, OpenCV, and geospatial tools. Experienced in deploying solutions on embedded platforms (Jetson), WebGIS, and cloud environments for real-world engineering and infrastructure applications.

Skills

Core Expertise

- Computer Vision, Deep Learning (CNN, YOLO)
- Photogrammetry, SfM/MVS, 3D Reconstruction, data visualization
- Geospatial AI, UAV & Remote Sensing

Programming & ML

- Python, C++, PyTorch, OpenCV
- CUDA, Linux, Docker, Git

Geospatial & Sensors

- GNSS (RTK/PPK), RTKLIB
- GIS (QGIS, GeoServer, WebGIS)
- LiDAR, Raster Analysis

Systems & Video

- FFmpeg, GStreamer, RTSP
- ROS, MAVLink
- Web APIs, Microservices

Experience

GEO-SUP - Computer Vision & Geospatial Engineer | Tehran, Iran | 2018 - PRESENT

- Delivered UAV photogrammetry and GNSS-integrated workflows for mining, terrain modeling, and infrastructure monitoring using SfM/MVS and PPK/RTK techniques.
- Built deep learning–based object detection and defect recognition systems (YOLO) for pavement analysis, vehicle classification, and parking detection.
- Processed large-scale LiDAR and image datasets for terrain and urban 3D modeling, including dense point clouds and mesh reconstruction.
- Developed custom QGIS extensions and integrated GIS workflows into operational decision-support systems.
- Implemented real-time video processing and stabilization pipelines using Python, C++, OpenCV, FFmpeg, and GStreamer.
- Deployed computer vision models on NVIDIA Jetson platforms for edge inference in field environments.

Navatel - SGAS -Full Stack Web Developer (Remote) | Tehran, Iran | 2021 - 2022

- Developed and maintained scalable web applications and APIs supporting real-time data ingestion and visualization.
- Implemented WebSocket-based services and microservices architectures for live data streaming and monitoring.
- Collaborated on system integration, API documentation (Swagger), and deployment using Docker.

CRPLab - Research Assistant | Tehran, Iran | 2015 - 2016

- Assisted in data acquisition and testing for UAV-related research
- Supported deployment of image-processing pipelines in field conditions
- Implemented camera calibration, feature extraction, and multi-view geometry workflows

Education

MSc Photogrammetry Engineering | School of Surveying and Geospatial Engineering| University of Tehran | Tehran, Iran | 2021 - 2024

- Focus on deep learning for remote sensing and image-based change detection.
- Teaching Assistant: Advanced Digital Image Processing

key courses:

- GIS / Remote Sensing
- Spatial Modeling
- Simulation / UAV photogrammetry
- Digital Twins from NeRF-Based 3D Reconstruction
- Data Analysis / Machine Learning

BSc – Geospatial Engineering | School of Surveying and Geospatial Engineering | University of Tehran | Tehran, Iran | 2015 -2020

- Proficiency in Remote Sensing, Geomatics, and GNSS GPS
- Advanced knowledge of Photogrammetry and Hydrology
- Competence in MATLAB, Bundle Adjustment, and Image Processing

Publication

Prediction of monthly dry days of Lake Urmia with Deep learning algorithms
In progress

Languages

English - Good proficiency

German - Basic knowledge

Arabic – Elementary

Projects

GNSS PPK Data Loggers: Developed GNSS PPK data loggers using Raspberry Pi, enhancing precision in navigation systems.

WebGIS for Parking: Developed a WebGIS application to assist in locating parking lots across Tehran, improving urban mobility.

Network sniffer application, IMDb Simulation Website: MCI Internship using Java Spring Boot, C++, MongoDB, Redis

Sparse Bundle Adjustment: Implemented robust Sparse Bundle Adjustment with the Levenberg-Marquardt method on UAV images using Python, enhancing image reconstruction accuracy.

Mine Positioning: Conducted static positioning for Gol Gohar mine's pit number one using multiple 2-frequency receivers.

Volume Calculation: Calculated monthly extracted soil volume using drone photogrammetry and 3D modeling, streamlining resource management.

Visualization: Integrated UAV photogrammetry and environmental datasets for high-resolution spatial modeling and visualization

Time-series forecasting: using remote sensing data

References

Associate Professor Dr. Mahdi Hasanlou | School of Surveying and Geospatial Engineering, University of Tehran | hasanlou@ut.ac.ir

Associate Professor Dr. Mohammad Saadat Seresht | School of Surveying and Geospatial Engineering, University of Tehran | msaadat@ut.ac.ir

Professor Arash Malekian | College of Agriculture and Natural Resources / Reclamation of Arid and Mountainous Regions | malekian@ut.ac.ir

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